

About document

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What is Spec-Kit?

- SpecKit is the tool open source of GitHub based on **Spec-Driven Development (SDD)**.
- The method to build application based on principles **Design before Implement**.
- AI reads the specification and generates code from it - not the other way around.

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Project structure

```
.specify/  
  memory/  
    constitution.md          ← Principles project  
  templates/                ← template for  
constitution/spec/plan/tasks/checklist  
specs/  
  001-feature-name/  
    spec.md                  ← Specification context  
    plan.md                  ← Plan technical  
    tasks.md                 ← List tasks  
    data-model.md  
    api-spec.json  
    research.md  
  002-feature-name/  
    spec.md                  ← Specification context  
    plan.md                  ← Plan technical  
    tasks.md                 ← List tasks
```

```
data-model.md
api-spec.json
research.md
```

Core project

Constitution

- This is the constitution of project, **Only a file constitution.md** in project.
- Set all principles for AI agent must be followed.
- AI read it first then implement features.

It contains:

- Standard code.
- Unit Test (TDD, coverage, ...)
- Principles UX/accessibility
- Constraint performance, security

The Cycle for a feature:

```
specify → clarify → plan → tasks → implement
```

- Each feature is built based on this cycle.
- Each feature runs this cycle independently.

In Specify - Specification

- Pure business description from the **user's point of view**, written as **User Stories**.
- Describe **What** (what the user wants) and **Why** (why they need it).
- Define user roles and real-world use cases.
- Describe what the user **expects to receive** after each action.

Rules:

-  Focus on user behavior and expectations.
-  No technology stack (React, NextJS, API...).
-  No description of how the system processes internally - that belongs to Plan.

Output: **spec.md**

In Clarify - Clarification

- Validate the **completeness of the Spec** before moving to technical planning.

- AI asks questions back to the user about unclear or missing points in `spec.md`.
- Surface **edge cases** not yet covered.
- Ensure AI and user reach **agreement on scope** before investing in Plan and Tasks.

Why it matters:

- Any ambiguity left in Spec will be **amplified** across Plan → Tasks → Implementation.
- Skipping this step leads to rework and wasted effort.

In Plan - Technical Architecture

- The **translation layer** from business language (Spec) into technical language (Code).
- This is where technology decisions are made.

It contains:

- `plan.md` - Tech stack, overall architecture, key technical decisions.
- `data-model.md` - Database schema, relationships between entities.
- `api-spec.json` - API endpoints, request/response contracts.
- `research.md` - Library evaluations, alternatives considered.

Output: `plan.md`, `data-model.md`, `api-spec.json`, `research.md`

In Tasks - Execution Roadmap

- Detailed execution plan generated from Spec + Plan.
- Each task is linked to a specific **User Story** from `spec.md`.

It contains:

- Task dependency order (Task B must follow Task A).
- Parallel tasks marked with `[P]` - can run simultaneously.
- Specific file paths to be created or modified.
- Definition of done for each task.

Output: `tasks.md`

Analyst

- An optional research phase that runs before Specify when the problem space is unclear.
- Used to explore the domain, research competitors, or validate ideas before writing a spec.
- Helps the user define what to build before committing to a User Story.

It contains:

- `research.md` - Market research, competitor analysis, feasibility notes.
- Key insights that inform the Spec (user needs, pain points, opportunities).

When to use:

- Starting a brand new project with no clear direction.
- Validating whether a feature is worth building.
- Exploring technical or business unknowns before specifying.

Rules:

-  Output feeds directly into spec.md the Specify phase.
-  No implementation decisions here — that belongs to Plan.

In Implement - Build

- AI reads `tasks.md` and builds step by step in the correct order.
- AI always references `constitution.md` throughout the entire implementation.
- Follow TDD if required by constitution.

Quick Start

```
# Install uv (Windows)
powershell -ExecutionPolicy ByPass -c "irm
https://astral.sh/uv/install.ps1 | iex"

# Initialize project
uvx --from git+https://github.com/github/spec-kit.git specify init
<project-name>

# Navigate and open in VS Code
cd <project-name>
code .
```

Proof Training

○ doit_speckit

ing App - Initial Page Setup >  ## Phase 3.1: Setup

o dependencies)

```
ponents in `components/ui/`
ilities in `src/lib/dateUtils.ts`
th pastel colors in `tailwind.config.ts`
```

```
tatus enum in `src/types/goal.ts`  
ctions in `src/lib/storage.ts`  
es in `src/lib/dateUtils.ts`
```

```

  in `src/components/GoalForm.tsx`
  in `src/components/GoalList.tsx`

```

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GitHub Spec Kit Tutorials

Net Ninja - 1/9

▶  Up & Running with GitHub Spec Kit #1 - Intro & Setup
Net Ninja

2  Up & Running with GitHub Spec Kit #2 - The /constitution...
Net Ninja

3  Up & Running with GitHub Spec Kit #3 - The /specify Command
Net Ninja

4  Up & Running with GitHub Spec Kit #4 - The /clarify Command
Net Ninja

5  Up & Running with GitHub Spec Kit #5 - The /plan Command
Net Ninja


 Up & Running with GitHub Spec
 Kit #6 - The /tasks Command
 Net Ninja

7  Up & Running with GitHub Spec Kit #7 - The /analyze Command
Net Ninja


 Up & Running with GitHub Spec Kit #8 - Implementing Tasks
 Net Ninja

9  Up & Running with GitHub Spec Kit #9 - Adding More Specs
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Tất cả

Trong loạt video

Của Net Ninja